Scales

1 Convert the following descriptions to a scale ratio:

a $3\ \text{cm}$ represents an actual distance of $9\ \text{m}$.

b $20\ \text{cm}$ represents an actual distance of $16\ \text{km}$.

c $8\ \text{mm}$ represents an actual distance of $40\ \text{m}$.

d $60\ \text{mm}$ represents an actual distance of $5400\ \text{m}$.

e $5\ \text{cm}$ represents an actual distance of $30\ \text{km}$.

f $98\ \text{mm}$ represents an actual distance of $9.8\ \text{km}$.

2 The scale on a map of a garden is $1 : 2000$. How far apart on the map should two fountains be drawn if the actual distance between the fountains is $100\ \text{m}$? Express your answer in centimetres.

3 The scale on a map is $1 : 400\,000$. How far apart on the map should two train stations be drawn if the actual distance between the stations is $100\ \text{km}$? Express your answer in centimetres.

4 Consider a map with a scale of $1 : 2000$.

a Find the actual distance (in centimetres) between two points which are drawn $19\ \text{cm}$ apart on the map.

b Find the distance between the two points in metres.

5 Consider a map with a scale of $1 : 25\,000$.

a Find the actual distance (in centimetres) between two points which are drawn $14\ \text{cm}$ apart on the map.

b Find the distance between the two points in kilometres.

6 Bianca is looking over a map of her local area and notices that the scale of the map is given as $1 : 100$ in the map legend.

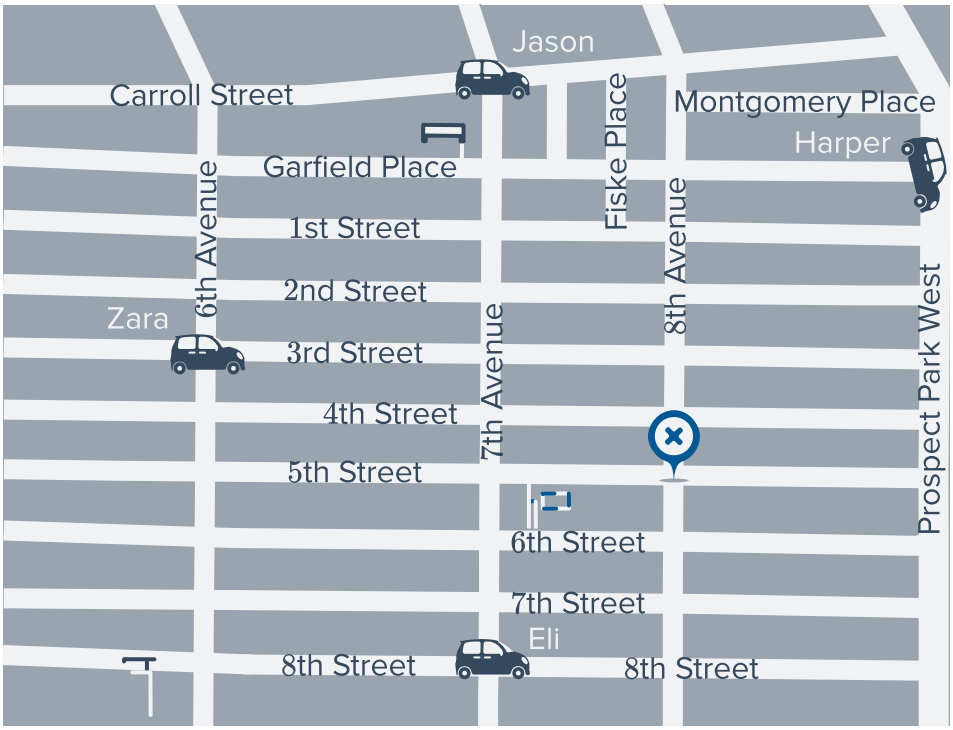
a Find the actual distance (in centimetres) between two points which are drawn $12\ \text{cm}$ apart on that map.

b Find the distance between the two points in metres.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Maps and scale diagrams

7 Below is a map of New York. The actual dimensions of one rectangular block is $230\,\text{m}$ by $80\,\text{m}$.



a How many metres will Zara need to drive to reach the pick-up point, marked with an x?

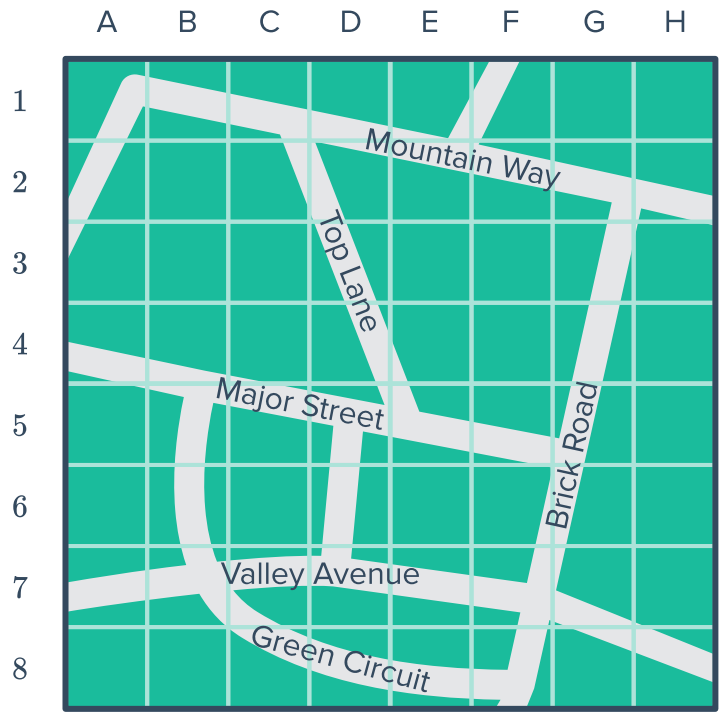
b Whose car is the closest to the pick-up location?

8 Use the map below to find the distance between Adelaide and Brisbane:



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

9 This is a map of where Adam and Carl live. The side of a small square on the map represents $100\,\text{m}$:



Adam is on the corner of Top Lane and Major Street (map reference $E5$). He walks $300\,\text{m}$ along Top Lane to Carl's house.

Write the map reference of Carl's house.

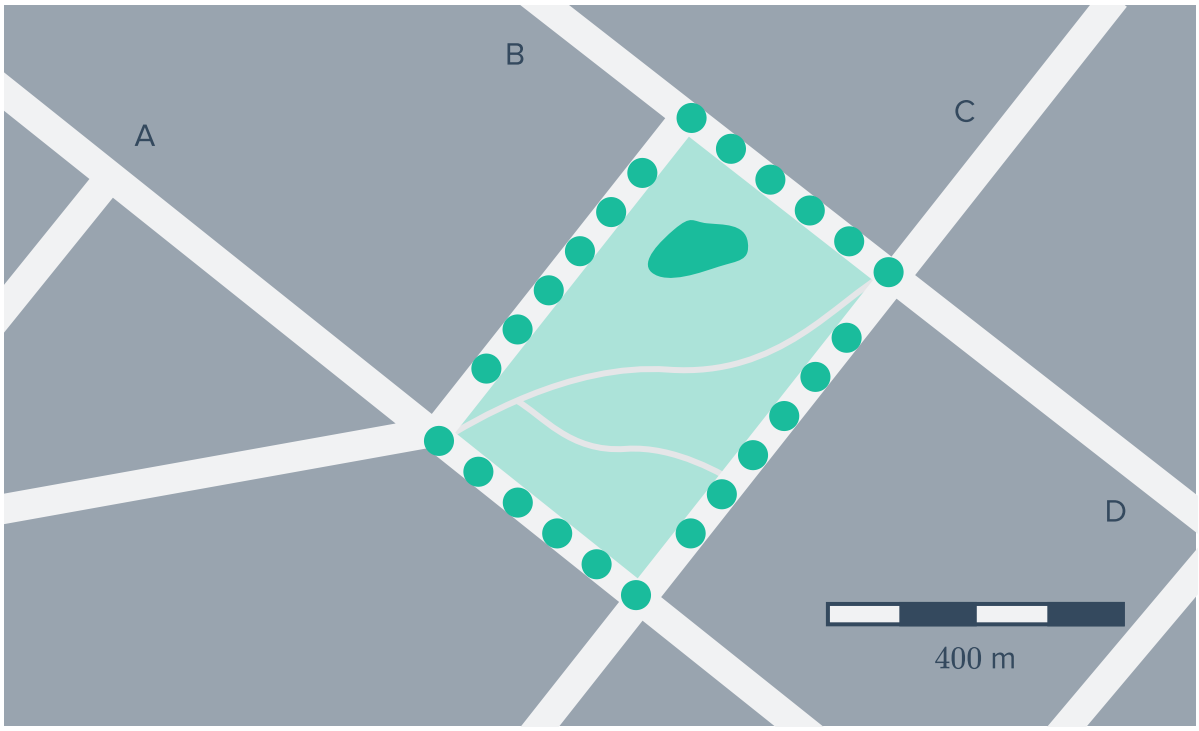
10 The map designer for a new amusement park measures the main street to be $4\,\text{cm}$. The walk along the main street is actually $120\,\text{m}$:



Write down the scale ratio of the map.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

11 A map of a town is drawn to scale below:



a Find the distance between house A and the park.

b Find the distance between house C and the park.

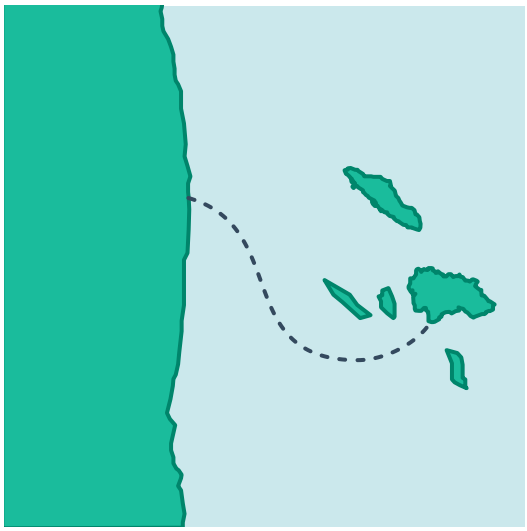
c Find the length of the shorter side of the park.

d Find the length of the longer side of the park.

e Find the distance traveled by driving from house A to house C along the roads.

f Find the distance traveled by first driving from house A to house C, then driving to house B, traveling along the roads.

12 The following is a $1 : 58\,000$ scale drawing of the sailing route from the mainland to an island off the coast. The captain approximates the distance to be 15.3 cm on the map. Estimate the distance in kilometres of the boat trip.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

13 Estimate the distance from Perth to Brisbane using the map scale:



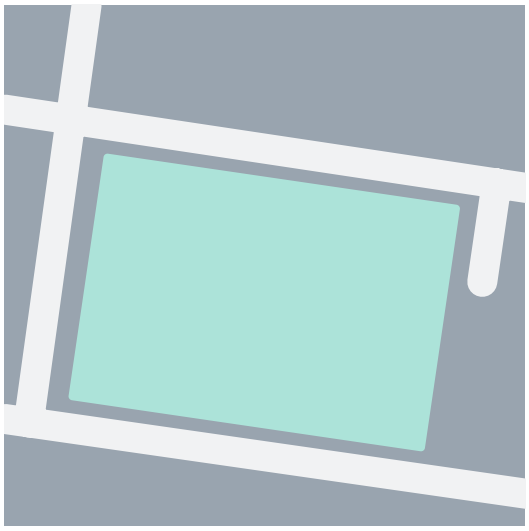
14 The scale on the map of a public garden is $1 : 5000$. How far apart on the map should two ponds be if the actual distance between the ponds is $350\ \text{m}$? Give your answer in centimetres.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

15 Vincent is comparing three parks from three different maps shown below. Which park has the largest area? Explain your answer.

Lesse park



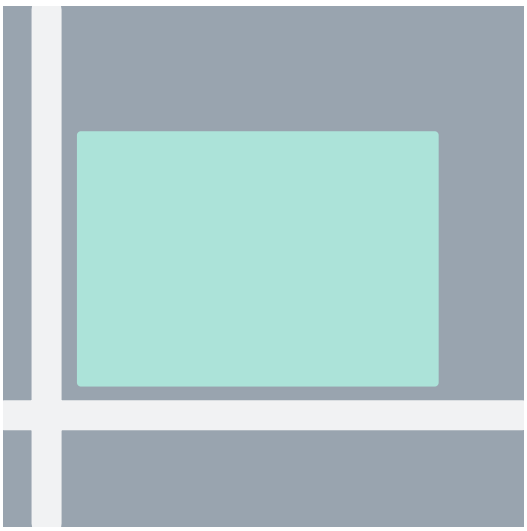
Scale: $1 : 2000$

Millenial park



Scale: $1 : 5000$

Tibsen park

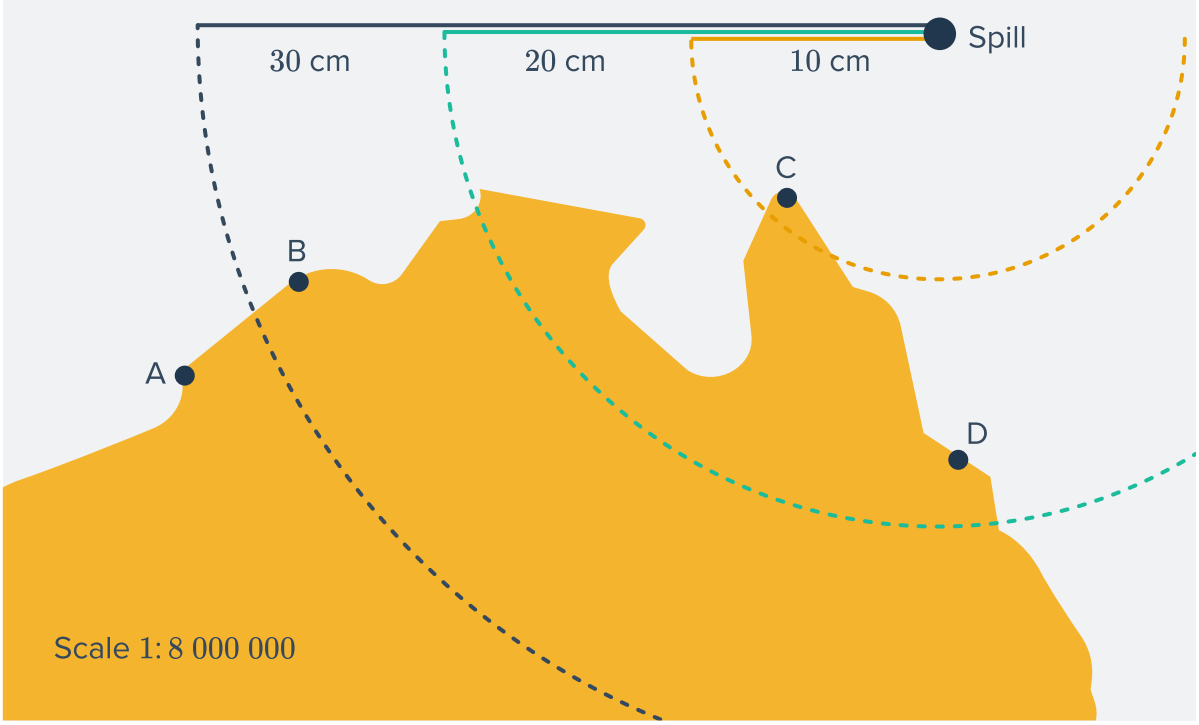


Scale: $1 : 7000$

16 A map is being drawn for a medium-sized rural town. The map will fit on an A4 piece of paper which has the dimension of $21\,\text{cm}$ by $29.7\,\text{cm}$. Write an appropriate scale for the map.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

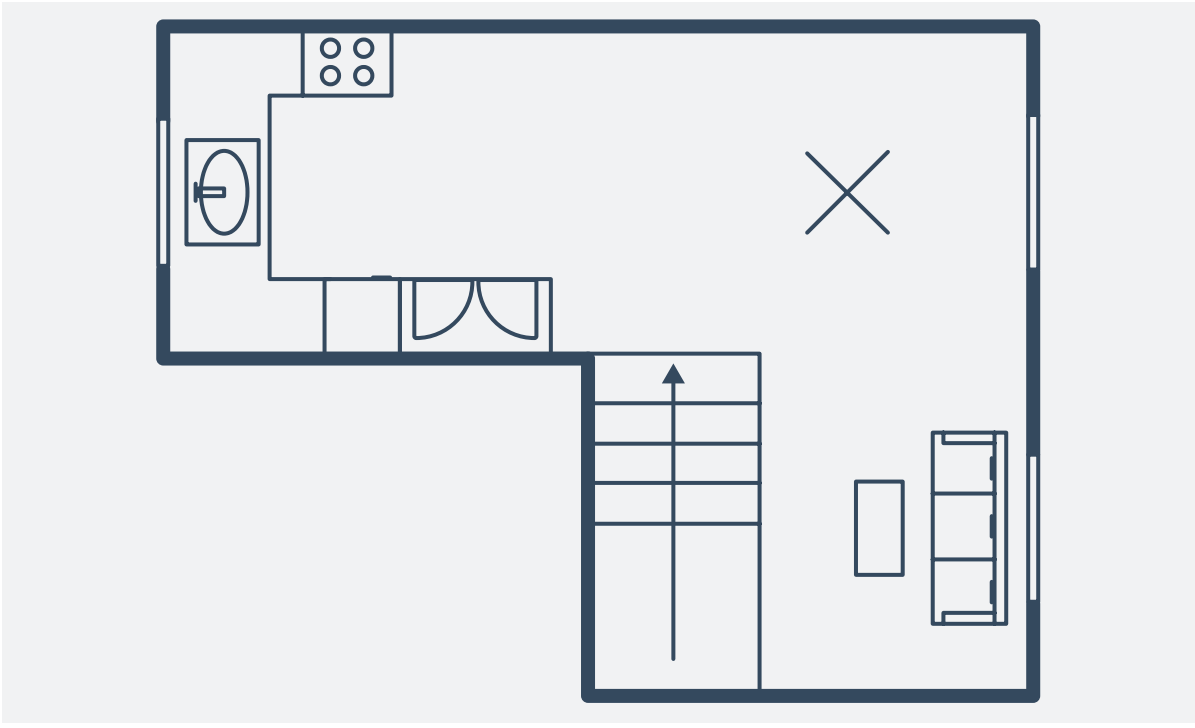
17 An oil tanker has a spill off the coast of Australia. An expert says that the oil could travel $1700\ \text{km}$ in the first day. Their map is shown sketched with circles with radius $10\ \text{cm}$, $20\ \text{cm}$ and $30\ \text{cm}$:



a Which locations will be affected in the first day?

b Location B is $26\ \text{cm}$ from the oil spill on the map. Find the actual distance of this location from the oil spill.

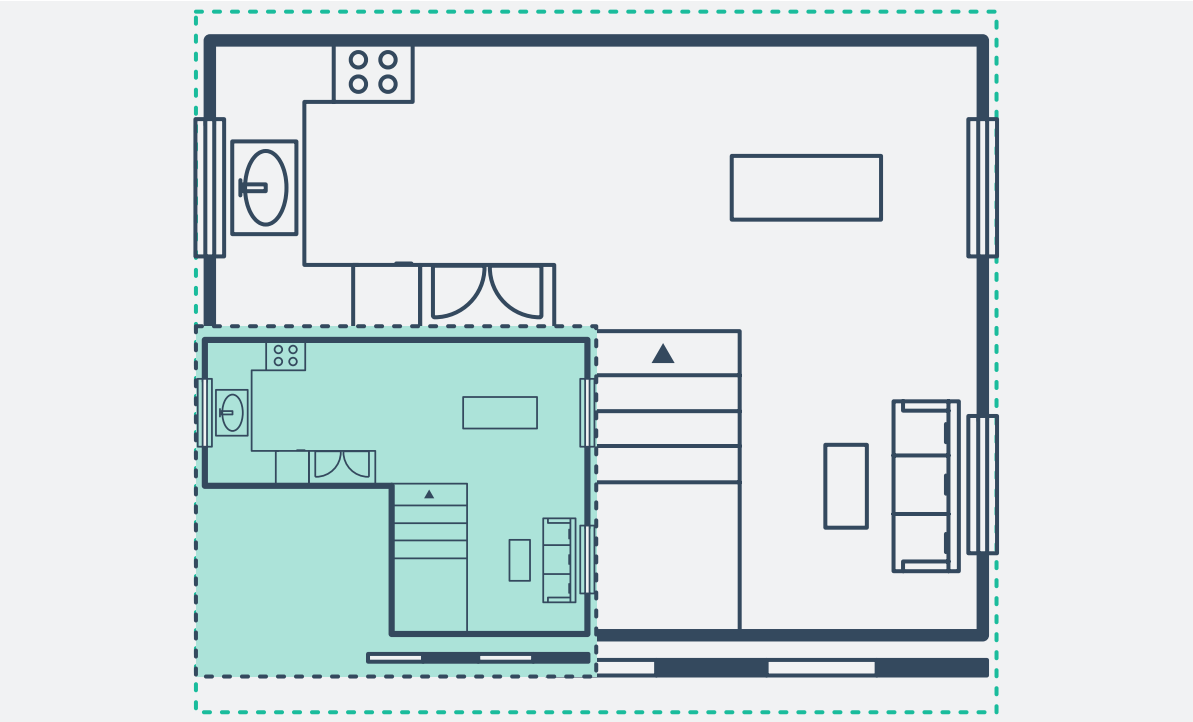
18 The following is a $1 : 200$ floor plan of a house. The homeowner wishes to add a dining room table, which is $150\ \text{cm}$ long, placing it where the $\times$ is marked on the floor plan.



Find the table's length that should be drawn to in the floor plans

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

19 A plan of a building with ratio $1 : 90$ is enlarged so its length and width are both doubled as shown:



a Find the new scale ratio for the larger plan.

b The scale bar on the original plan represents $5\ \text{m}$. When the plan is doubled, the scale bar is doubled as well. Find the length that the new scale bar represents.

20 An architect is designing a house that is $21\ \text{m}$ long and $15\ \text{m}$ wide.

a She wants to print the plans for this house in a $84\ \text{cm}$ by $60\ \text{cm}$ section of a large piece of paper. Find the ratio she should use to fit the plans on the paper.

b After printing the plans, the architect measures the length of the hallway to be $24\ \text{cm}$ on paper. She wants to order a rug that covers the full length of the hallway.

How long should the rug be in metres?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647